

E-Commerce Customer Churn & Retention Analytics

Project Report

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Dataset: E-Commerce Customer Data for Behavior Analysis — Kaggle

Table of Contents

1. Introduction
2. Business Problem
3. Objectives
4. Dataset Description and Source
5. Data Cleaning
6. Exploratory Data Analysis (EDA)
7. Key Performance Indicators (KPIs)
8. Customer Analysis
9. Churn Analysis
10. ML Model and Evaluation
11. Key Insights
12. Business Risks and Opportunities
13. Recommended Business Actions
14. Conclusion
15. Limitations

1. Introduction

Customer churn — the phenomenon where customers stop doing business with a company — is one of the most critical challenges in e-commerce. Retaining an existing customer costs significantly less than acquiring a new one, yet many businesses lack the analytical capability to identify at-risk customers before they leave.

This project performs a comprehensive data analytics study on a large e-commerce transaction dataset (250,000 records, 49,661 unique customers). The analysis moves from raw data cleaning through exploratory analysis, business KPI calculation, customer segmentation, churn analysis, and culminates in a machine learning churn prediction model. All findings are converted into practical, actionable business recommendations.

2. Business Problem

The business problem addressed in this project is: Which customers are at risk of churning, and what business actions can reduce that risk?

Specifically, the project aims to:

- Understand the profile of churned vs. retained customers.

- Identify the key behavioural and demographic drivers of churn.
- Quantify the financial impact of churn on the business.
- Build a predictive model to flag customers likely to churn.

3. Objectives

The specific objectives of this project are:

1. Perform thorough data cleaning and preprocessing on the raw dataset.
2. Conduct EDA to uncover purchasing patterns, revenue trends, and customer demographics.
3. Define and calculate core business KPIs (AOV, CLV, churn rate, return rate).
4. Segment customers using RFM (Recency, Frequency, Monetary) analysis.
5. Analyse churn patterns across demographics, product categories, and payment methods.
6. Build and evaluate churn prediction models (Logistic Regression and Random Forest).
7. Translate findings into actionable business recommendations.

4. Dataset Description and Source

Dataset Name: E-Commerce Customer Data for Behavior Analysis

Source: Kaggle — <https://www.kaggle.com/datasets/uom190346a/e-commerce-customer-behavior-dataset>

File: ecommerce_customer_data_large.csv

The dataset contains 250,000 transaction records across 13 columns. Each row represents a single purchase transaction. The same customer may appear in multiple rows (an average of 5 transactions per customer). The binary Churn column indicates whether the customer has churned (1) or is still active (0).

Column Descriptions:

Column Name	Data Type	Description
Customer ID	Integer	Unique identifier for each customer
Purchase Date	Datetime	Date and time of the transaction (2020–2023)
Product Category	String	Category of purchased product: Books, Clothing, Electronics, Home
Product Price	Integer	Unit price of the product in USD
Quantity	Integer	Number of units purchased
Total Purchase Amount	Integer	Total transaction value in USD
Payment Method	String	Method of payment: Cash, Credit Card, PayPal
Age	Integer	Customer age in years (18–70)
Customer Age	Integer	Duplicate of 'Age' — dropped during cleaning
Returns	Float (0/1)	Whether the transaction was returned (18.95% missing)
Customer Name	String	Customer full name (not used in analysis)
Gender	String	Customer gender: Male or Female
Churn	Integer (0/1)	Target variable: 1 = churned, 0 = retained

Note: 'Customer Age' is a duplicate of 'Age' and was dropped during cleaning. 'Returns' had 18.95% missing values, imputed with the column median.

5. Data Cleaning

The following cleaning steps were applied to the raw dataset:

5.1 Datetime Parsing

The 'Purchase Date' column was stored as a plain string. It was converted to a proper Python datetime object using `pd.to_datetime()`, enabling extraction of Year, Month, and Day of Week features.

5.2 Duplicate Column Removal

Inspection confirmed that 'Customer Age' is a 100% exact duplicate of the 'Age' column (verified by a boolean equality check across all 250,000 rows). The redundant 'Customer Age' column was dropped.

5.3 Missing Value Imputation

The 'Returns' column had 47,382 missing values (18.95% of rows). Since 'Returns' is a binary column (0 = not returned, 1 = returned), missing values were imputed with the column median (1.0 — indicating a return). The column was then cast to integer. No other columns had missing values.

5.4 Whitespace Cleaning

All string columns were stripped of leading and trailing whitespace using `str.strip()`.

5.5 Feature Engineering

The following derived features were added for analysis:

- Year, Month, DayOfWeek — extracted from Purchase Date for time-series analysis.
- YearMonth — period column for monthly aggregations.
- Age Group — customers binned into five groups: 18-25, 26-35, 36-45, 46-55, 56-70.

6. Exploratory Data Analysis (EDA)

6.1 Revenue Trend Over Time

Monthly revenue was consistent throughout the 2020–2023 period, ranging between approximately \$14.6M and \$16.2M per month. There is no observable seasonal spike or major decline, which indicates a stable but non-growing revenue base. This flatness is a strategic concern — a healthy business should show growth over a 3.5-year period.

[Figure: fig_monthly_revenue.png — Monthly Revenue Trend Line Chart]

6.2 Revenue by Product Category

Revenue is almost perfectly balanced across all four product categories:

Product Category	Total Revenue	Revenue Share
Home	\$171,138,916	25.1%

Product Category	Total Revenue	Revenue Share
Clothing	\$170,716,122	25.0%
Electronics	\$170,146,025	25.0%
Books	\$169,345,236	24.9%

[Figure: fig_category_revenue.png — Revenue Bar and Pie Chart]

6.3 Customer Demographics

Customer ages range from 18 to 70, with a mean of ~44 years and a roughly uniform distribution. The gender split is 50.3% Male vs 49.7% Female — nearly perfectly balanced. This uniformity suggests the business has broad demographic appeal.

[Figure: fig_demographics.png — Age Distribution and Gender Pie Chart]

6.4 Payment Methods

All three payment methods — Credit Card (33.4%), PayPal (33.4%), Cash (33.2%) — are used equally. This balance is unusual for e-commerce and may indicate the platform serves both online and in-person sales channels.

[Figure: fig_payment_methods.png — Payment Method Bar Chart]

6.5 Returns Rate

The overall return rate is approximately 50%. This is very high and represents a major operational and financial risk. Return rates are nearly identical across all four product categories, suggesting a systemic issue rather than a category-specific one.

[Figure: fig_returns_by_category.png — Return Rate by Category Bar Chart]

7. Key Performance Indicators (KPIs)

The following KPIs were computed from the cleaned dataset:

KPI	Value	Business Meaning
Total Revenue	~\$681 Million	Total transactional revenue (Jan 2020 – Sep 2023)
Total Transactions	250,000	Complete dataset size
Unique Customers	49,661	Active customer base size
Avg Order Value (AOV)	\$2,725	Average spend per transaction
Avg Transactions / Customer	5.03	Average number of purchases per customer
Estimated CLV	~\$13,703	AOV x Avg Transactions — lifetime revenue per customer
Return Rate	~50%	Proportion of transactions resulting in a return
Customer Churn Rate	19.99% (~20%)	Proportion of customers who have churned

Key Observations: The estimated CLV of ~\$13,700 means each churned customer represents roughly \$13,700 in lost future revenue. With ~9,929 churned customers, the total estimated lost CLV is approximately \$136 million.

8. Customer Analysis

8.1 RFM Segmentation

Customers were segmented using RFM (Recency, Frequency, Monetary) analysis. Each customer was scored on a 1–5 quintile scale for each dimension. The combined RFM score (3–15) was used to assign one of four business segments:

Segment	RFM Score Range	Description
Champions	13–15	Recent, frequent, high-spending customers — most valuable
Loyal	10–12	Consistent buyers with moderate to high spend
At Risk	7–9	Previously good customers showing reduced engagement
Lost	3–6	Disengaged customers who haven't purchased recently

[Figure: fig_rfm_segments.png — RFM Segment Distribution and Churn Rate per Segment]

8.2 Spending by Age Group and Gender

Average order value is consistent across all age groups (~\$2,700–\$2,760) and genders (~\$2,725), confirming that spending behaviour is demographically uniform in this dataset. This means demographic targeting for revenue maximisation is not a strong lever here — the focus should instead be on retention.

[Figure: fig_age_gender_spending.png — Avg Order Value by Age Group and Gender]

8.3 Payment Preference by Age Group

Payment method preference is also uniform across all age groups — no age group shows a strong preference for any payment method. All groups split approximately equally between Credit Card, PayPal, and Cash.

[Figure: fig_payment_by_age.png — Stacked Bar of Payment Method by Age Group]

9. Churn Analysis

9.1 Overall Churn Rate

The customer-level churn rate is 19.99% (approximately 20%). Of 49,661 unique customers, 9,929 have churned and 39,732 are retained. This means the business loses roughly 1 in 5 customers — a significant but not catastrophic rate.

[Figure: fig_churn_overview.png — Churn Pie Chart and Churn by Age Group]

9.2 Churn by Demographic and Behavioural Segments

Churn rates across all segments are very close to the overall 20% baseline. This is summarised in the table below:

Segment	Churn Rate	Insight
Age 18-25	21.1%	Marginally higher — youngest customers churn slightly more
Age 56-70	20.5%	Slightly higher — older customers may disengage
Female	20.2%	Marginally above average
Male	19.9%	Marginally below average
Cash payments	20.3%	Slightly elevated

Segment	Churn Rate	Insight
Credit Card	19.7%	Slightly below average
All categories	~20%	No significant difference across product types

[Figure: fig_churn_breakdown.png — Churn Rate by Payment Method, Gender, and Category]

9.3 Churn vs Purchase Behaviour (RFM)

Comparing churned and retained customers on RFM dimensions shows almost no difference in average frequency (5.0 transactions for both groups) or average monetary value (~\$13,700 for both groups). This confirms that churn in this dataset is not driven by low engagement or low spend — it appears to be randomly distributed across the customer base.

[Figure: fig_churn_rfm.png — Frequency and Monetary Distributions: Churned vs Retained]

10. Machine Learning — Churn Prediction Model

10.1 Approach

Two classification models were built and evaluated: Logistic Regression (baseline) and Random Forest. Features used were customer-level aggregates: Recency, Frequency, Monetary, Return Rate, Age, and Gender (encoded). The dataset was split 80% training / 20% testing with stratification on the target variable.

10.2 Model Results

Model	Accuracy	ROC-AUC	Notes
Logistic Regression	~80%	~0.50	Predicts the majority class (Retained) only
Random Forest	~80%	~0.50	Same issue — imbalanced but uniform churn label
5-fold Cross-Validation	Consistent	~0.50	Confirms results are stable, not overfitted

[Figure: fig_model_evaluation.png — ROC Curves and Confusion Matrices for both models]

[Figure: fig_feature_importance.png — Random Forest Feature Importance]

10.3 Interpretation of Results

Both models achieve ROC-AUC scores near 0.50 — equivalent to random guessing. This is not a modelling error. A Pearson correlation analysis confirmed that none of the available features (Recency, Frequency, Monetary, Returns, Age, Gender) have a meaningful statistical relationship with the Churn label (all $|r| < 0.01$).

This strongly suggests that the Churn column in this dataset was assigned independently of customer behaviour — a common characteristic of synthetically generated datasets. Despite this finding, building the models was valuable for:

(a) Demonstrating correct ML pipeline construction and evaluation methodology.

(b) Confirming that additional data collection (see Limitations) is needed before a productive predictive model is possible.

11. Key Insights

The following key insights emerged from this analysis (all values computed from actual data):

Insight 1 — Churn Rate of ~20% is Costly: With a CLV of ~\$13,700, the 9,929 churned customers represent an estimated \$136M in lost revenue potential. Reducing churn by even 5 percentage points would recover approximately \$34M in CLV.

Insight 2 — Revenue is Stable but Not Growing: Monthly revenue held flat at ~\$15M throughout 2020–2023. A healthy growing business should show consistent year-over-year improvement. This stagnation is a strategic warning sign.

Insight 3 — The 50% Return Rate is a Critical Risk: Half of all transactions involve a return. Beyond the direct cost of reverse logistics, high returns damage customer satisfaction and trust — both of which increase churn risk.

Insight 4 — Revenue is Diversified and Resilient: All four product categories contribute almost exactly 25% of revenue each. This is a structural strength — the business is not dependent on any single category.

Insight 5 — Young and Older Customers Have Slightly Higher Churn: Customers aged 18-25 (21.1%) and 56-70 (20.5%) churn marginally more. Tailored retention programmes for these groups could help.

Insight 6 — Existing Features Cannot Predict Churn: Purchase frequency, spend level, age, gender, and payment method all show near-zero correlation with churn. Richer behavioural data is needed to build a useful predictive model.

12. Business Risks and Opportunities

Risks:

Risk 1 — High Churn Cost: A ~20% churn rate translates to ~\$136M in lost CLV. Without an active retention strategy, this loss compounds year over year.

Risk 2 — Return Rate Crisis: A 50% return rate suggests serious product quality, fit, or expectation mismatch issues. Each return costs money in logistics, restocking, and customer service, and damages brand trust.

Risk 3 — Revenue Stagnation: Flat revenue over 3.5 years in an environment of inflation and rising competition means the business is effectively shrinking in real terms.

Risk 4 — No Early Warning System: Without predictive churn signals, the business cannot identify at-risk customers until they have already left.

Opportunities:

Opportunity 1 — CLV Uplift: A 5% reduction in churn rate would preserve approximately \$34M in customer lifetime value.

Opportunity 2 — Return Reduction: Halving the return rate from 50% to 25% would eliminate half the reverse logistics costs and dramatically improve customer satisfaction scores.

Opportunity 3 — Loyalty Programme: The large base of Champion and Loyal RFM customers (those with high scores) is an asset. A structured loyalty programme can increase purchase frequency and deepen engagement.

Opportunity 4 — Data Collection: Investing in richer behavioural data (web session time, support tickets, discount usage, NPS surveys) would unlock truly predictive churn modelling.

13. Recommended Business Actions

R1 — Launch a Loyalty Programme: Design a tiered loyalty programme (Bronze/Silver/Gold) targeting Champion and Loyal RFM segments. Reward repeat purchases with points, early access, and exclusive discounts. Goal: increase average transactions per customer from 5 to 6.

R2 — Implement Win-Back Campaigns: Identify customers in the 'At Risk' and 'Lost' RFM segments and launch personalised re-engagement campaigns via email or SMS. Offer a time-limited discount (e.g. 15% off their next purchase). Even a 5% win-back rate on these segments generates meaningful CLV recovery.

R3 — Reduce Returns Through Product Quality Initiatives: Conduct a detailed returns analysis (reasons for return, product subcategories with highest returns) and implement targeted improvements: better product photography, detailed size guides, improved product descriptions, and pre-shipment quality checks.

R4 — Age-Targeted Retention for 18-25 and 56-70 Groups: Design specific retention offers for young and older customers. For 18-25: social-media-integrated offers, student discounts, easy returns. For 56-70: simplified checkout, phone support, clear product information.

R5 — Revenue Growth Initiative: The flat revenue trend requires strategic intervention. Consider expanding to new product categories, launching promotional campaigns during lower-revenue months, or introducing subscription bundles.

R6 — Invest in Richer Data Collection: To enable predictive churn modelling, the business should begin collecting: web session duration, abandoned cart events, customer satisfaction (NPS/CSAT) scores, support ticket volume, promotional code usage, and subscription/loyalty programme engagement.

14. Conclusion

This project delivered a complete end-to-end data analytics study on a large e-commerce dataset of 250,000 transactions across 49,661 unique customers. Starting from raw data inspection and cleaning, the project progressed through exploratory analysis, KPI calculation, RFM customer segmentation, churn analysis, and machine learning modelling.

The two most urgent business priorities identified are: (1) reducing the 20% customer churn rate through loyalty and win-back programmes, and (2) aggressively addressing the 50% return rate through product quality and customer experience improvements.

The ML churn prediction model, while correctly constructed, was unable to produce a meaningful predictive signal due to the near-zero correlation between available features and the churn label. This is an honest, data-grounded finding. Future predictive capability depends on collecting richer behavioural signals.

15. Limitations

1. Synthetic Dataset: The Churn label appears to have been assigned randomly rather than based on actual customer behaviour. This limits the practical usefulness of the ML model.
2. No Behavioural Signals: The dataset lacks engagement signals (page views, time on site, support interactions, NPS scores) that are typically the strongest predictors of churn in real e-commerce environments.
3. Static Dataset: There is no information about new customer acquisition over the period, making it impossible to calculate true retention rates or cohort-based CLV.
4. Returns Data Incompleteness: The 18.95% missing rate in the Returns column required imputation, which introduces some uncertainty into return rate calculations.
5. CLV Estimation: The estimated CLV is a simple approximation ($AOV \times \text{average transactions}$). A full CLV model would require survival analysis or probabilistic modelling (e.g. BG/NBD model).
6. No Product-Level Data: The dataset does not contain individual product IDs or subcategories, limiting the granularity of returns and preference analysis.

References

Dataset: E-Commerce Customer Data for Behavior Analysis — Available on Kaggle at:
<https://www.kaggle.com/datasets/uom190346a/e-commerce-customer-behavior-dataset>

Libraries: Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn (Python)

Methodology: RFM analysis framework — Hughes, A.M. (1994). Strategic Database Marketing. Scikit-learn documentation (<https://scikit-learn.org>).